

L&T EduTech Project Submission

AI-Based Road Surface Defect Monitoring System

Day 5: Dashboard Integration & Algorithmic Logistics

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5.1 Front-End Web Portal & Dashboard Deployment

Day 5 culminated in the transformation of raw inference matrices into actionable urban planning intelligence. We built a unified web portal using the Gradio library. The backend of this portal hosts our Python YOLOv8 processing script, while the frontend hosts an embedded Google Looker Studio / Power BI dashboard connected directly to the live Google Sheets ledger generated on Day 4.

Key visual components deployed in the dashboard include:

- **Spatial Defect Mapping:** A geographical bubble map utilizing a concatenated 'Latitude, Longitude' calculated field to pinpoint defect clusters globally. Bubble size is driven by severity, and color designates priority.
- **Operational Scorecards:** Real-time dynamic counters tracking Total Anomalies Detected, Average Priority Score, and Platform Source efficiency (assessing Drone vs. Vehicle asset coverage).
- **Cross-Filtering Controls:** Enabling municipal engineers to instantly filter the entire map to show only "CRITICAL" priority issues detected in the last 24 hours.

5.2 Priority Scoring Matrix

To facilitate automated triage and remove human bias from maintenance planning, a Priority Scoring algorithm was coded into the backend. Baseline weights were empirically assigned: Severe Potholes (10), Severe Cracks (7), Mild Potholes (5), Mild Cracks (3).

The composite score calculation is given by:

$$Score = \sum (BaseWeight_i \times Confidence_i)$$

The system then translates this continuous float into categorical operational levels: Scores ≥ 8.0 trigger a **CRITICAL** alert, ≥ 4.0 trigger **HIGH**, and lower scores populate the **MEDIUM** and **LOW** preventative maintenance queues.

5.3 Time-Optimized Fleet Routing & Dispatch

Addressing the final milestone, we integrated a mathematical logistics framework for automated maintenance dispatch. Once a Critical defect populates the database, the system must assign it to the correct repair crew.

- **Greedy Nearest-Neighbor Logic:** The system utilizes the Haversine formula to compute true spherical geographical distances between active maintenance fleets (lat_1, lon_1) and the logged defect coordinates (lat_2, lon_2) :

$$d = 2r \arcsin \left(\sqrt{\sin^2 \left(\frac{lat_2 - lat_1}{2} \right) + \cos(lat_1) \cos(lat_2) \sin^2 \left(\frac{lon_2 - lon_1}{2} \right)} \right)$$

- **Dynamic Travel ETA:** Transit times are dynamically calculated assuming a constrained urban traffic velocity of $v \approx 20km/h$.
- **Variable Repair Durations:** The algorithm stacks crew schedules based on job severity profiles. For example, Severe Potholes append +3.5 hours to a crew's queue, while Mild Cracks add only +1.0 hour, ensuring overlapping shifts are avoided.

5.4 Conclusion & Smart City Impact

The fully integrated system—from raw YOLOv8 inference to the Haversine-optimized Power BI dispatch mapping—successfully satisfies all L&T EduTech project milestones. By transitioning from manual, reactive inspections to an automated, confidence-routed AI monitoring pipeline, municipalities gain total visibility over their infrastructure health. This data-driven approach allows for precise budget allocation, significantly reduces the exorbitant costs associated with emergency foundation failures, and actively secures urban transportation networks.